

1. An electrostatic field in the xy-plane is given by the potential function $\phi = 3x^2y - y^3$, find the stream function.

Ans. $\psi = 3xy^2 - x^3 + c$.

2. In a two-dimensional fluid flow, the stream function is $\tan^{-1} \frac{y}{x}$, find the velocity potential function.

Ans. $\frac{1}{2} \log(x^2 + y^2)$

3. A force field is given by $F = 3z + 5$. Find the work done in moving an object in this force field along the parabola $z = t^2 + it$ from $z = 0$ to $z = 4 + 2i$.

Ans. 50.

4. An object is displaced by a force field $F = 4z + 8$. Find the work done in displacing this object by force field F along the parabola $z = t^2 + it$ from $z = 0$ to $z = 9 + 3i$.

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5. Assume that $f(z)$ is analytic on the annulus (i.e.,) $A: r_1 < |z - z_0| < r_2$, then $f(z)$ is expressed in terms of series is:

$$F(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} b_n(z - z_0)^{-n}$$

where coefficients a_n and b_n are constants. From the series, we can also say,

- $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converge to the analytic function when $|z - z_0| < r_2$
- $\sum_{n=-1}^{\infty} b_n(z - z_0)^{-n}$ converge to the analytic function when $|z - z_0| > r_1$.

Thus, together both series converges on the annulus (A), where f is analytic.

Expand the function $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ in the following regions:

(I) $|z| < 2$ (II) $2 < |z| < 3$ (III) $|z| > 3$.

Ans. (I) $f(z) = 1 + \frac{3}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$

(II) $f(z) = 1 + \frac{3}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z}\right)^n - \frac{8}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{3}\right)^n$

$$(III) f(z) = 1 + \frac{3}{z} \sum_0^{\infty} (-1)^n \left(\frac{2}{z}\right)^n - \frac{8}{3} \sum_0^{\infty} (-1)^n \left(\frac{3}{z}\right)^n$$

6. If $f(z)$ is an analytic and its derivative $f'(z)$ is continuous within and on a simple closed curve C , then $\int_C f(z) dz = 0$. Based on the above case study, evaluate the following integral:

If $f(z) = \frac{2z^2+5}{(z+2)^3(z^2+4)}$, C is a square with vertices at $1+i, 2+i, 2+2i, 1+2i$.

If $f(z) = 2z^2 + 5z + c$, C is a square with vertices at $0+0i, 1+0i, 1+i, 0+i$.

7. A chemical engineer is studying the concentration of a certain substance in a reactor over time. The concentration $C(t)$ of the substance is given by the function $C(t) = \frac{1}{t^2 + 4t + 5}$, where t is the time in hours. The engineer wants to find the total amount of the substance that has reacted in the first 2 hours. Use the residue theorem to solve this problem.

8. A biologist is studying the growth of bacteria in a controlled environment. The growth rate of the bacteria is given by the function $G(z) = \frac{1}{z^2 + 2z + 1}$, where z is a complex variable representing time. The biologist wants to find the total amount of bacteria that have grown over the first 3 time units. Use the residue theorem to solve this problem.

9. A physics student is studying the behavior of a vibrating system. The displacement of the system is given by the function $d(t) = \frac{1}{t^2 + 2t + 2}$ where t represents time. The student wants to find the total energy of the system over the first 4 seconds. Use the residue theorem to solve this problem.

10. A finance student is analyzing the behavior of a stock market index. The daily change in the index is given by the function $I(z) = \frac{1}{z^2 + 3z + 2}$, where z is a complex variable representing trading days. The student wants to find the total change in the index over the first 5 trading days. Use the residue theorem to solve this problem.

11. An electrical engineer is designing a circuit with the transfer function $H(s) = \frac{1}{s^2 + 6s + 10}$, where s is a complex variable representing frequency in radians per second. The engineer wants to find the resonant frequency of the circuit. Use the residue theorem to solve this problem.